



Experimental and simulation-based reactor-scale strategies for engineering high mixotrophic synthesis, carbon sequestration and inter-trophic synergy in algal photobioreactors

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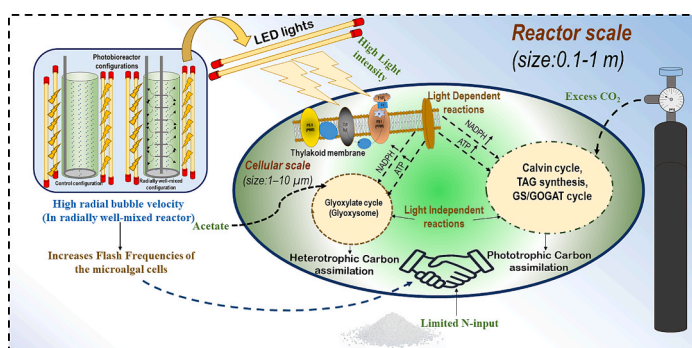
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HIGHLIGHTS

- This experimental-simulation study engineers high mixotrophic microalgal synthesis.
- Uses radial sparging, super-optimal illumination and CO₂, and nitrogen limitation.
- CFD analysis shows that radial mixing increases flashlight frequency by 3–5 fold.
- Flashlight effect enhances CO₂ sequestration (by 23–72%) and inter-trophic synergy.
- Deployable outdoors for remediation of acetate-rich water and CO₂-rich flue gases.

GRAPHICAL ABSTRACT



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ABSTRACT

This experimental and simulation study employs radial sparging – along with other reactor-scale parameters, such as illumination intensity and carbon (CO₂ and acetate) and nitrogen loading – to enhance carbon sequestration and inter-trophic synergy in mixotrophic cultivation of *Chlorella sorokiniana* in bubble-column photobioreactors. Radial mixing produces (3–5 times) higher flashlight frequencies, which enhance the synergy between phototrophy and heterotrophy and increase CO₂ sequestration by 23–72 % at 0.04–6 % CO₂ loading and 10,500–14,000 lx light. Lipid amounts obtained at 10,500, 12,000 and 14,000 lx peak at 3 %, 4 % and 6 % CO₂ to 1.61, 1.03, and 0.90 g/L, respectively, with corresponding inorganic carbon concentrations of 1.85, 1.40, and 1.49 g/L. A tetrad of reactor-scale engineering strategies – super-optimal CO₂ (6 %) and illumination (14,000 lx) with 12-hour light–dark circadian cycle, radial mixing and nitrogen limitation – in industrial-scale outdoor algal photobioreactors could sequester 23 % more CO₂ from flue-gases, with simultaneous bioremediation of acetate-rich wastewater.

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1. Introduction

Microalgal cultivation in enclosed photobioreactors (PBRs) and open raceway ponds, if scalable, could become a viable technology for carbon sequestration and renewable energy production (Alavianghavanini et al., 2024; Daneshvar et al., 2022). Photosynthetically efficient microalgae sequester atmospheric carbon dioxide (CO₂) more efficiently than higher plants by rapidly converting inorganic carbon to lipids, which could be *trans*-esterified to biodiesel (Bharti et al., 2024; Policastro et al., 2024). Microalgae also remove heavy metals and other toxic pollutants from wastewater (Mehta and Chakraborty, 2022). Industrial effluents and waste gases, such as CO₂ and sulfur oxides (SO_x), may be introduced into water bodies containing microalgae in several stages for bioremediation (Yen et al., 2015). These microalgae assimilate organic carbon and nitrogen sources from potential wastewater sources (Mehta and Chakraborty, 2022), along with sequestering CO₂ through carbon concentrating mechanism (at atmospheric conditions) and direct diffusional transport (for CO₂ > 1 %) (Raven et al., 2017; Nair and Chakraborty, 2020). CO₂-rich industrial flue gases, which when released into the atmosphere cause global warming and climate change (Sun et al., 2024), could be used for microalgal cultivation (Zhao et al., 2015). Thus, mass-scale microalgal cultivation in industrial reactors and raceway ponds is a technology that addresses a triad of environmental concerns – climate change, energy needs, and water security (Yin et al., 2022).

Microalgal photosynthesis in PBRs or ponds – from CO₂ sequestration to biomass production – has been deconstructed as a multiscale process occurring over disparate length scales (Mehta and Chakraborty, 2021). The macroscale is the reactor scale (size: 0.1–1 m), which is represented by variables such as overall biomass yield and manipulated through parameters such as radial mixing and light penetration. The mesoscale is the cellular scale (size: 1–10 µm), which is represented by cellular-level variables such as macromolecular content of lipids, proteins, and carbohydrates. The microscale is the organelle scale (~Å), which includes cell organelles such as mitochondria, glyoxysomes, and chloroplasts, where the photo-biochemical reactions occur. A sub-microscale may also be considered, where sub-proteins such as ferredoxin, plastocyanin, and plastoquinone molecules facilitate electron and proton transport. All these four structural scales are embedded hierarchically into each other such that inter-scale transport of mass and energy can produce a ‘whole greater than the sum of its parts’. Complex systems capable of self-regulation and energy-organization (such as photosynthesis) must necessarily contain intra- and inter-scale feedback loops. Here, we experimentally examine how manipulating macroscale parameters, such as reactor mixing, illumination and CO₂ and nitrogen loading, can affect the mesoscale synthesis of lipids, carbohydrates and proteins, and the elemental uptake and the synergy between phototrophic and heterotrophic pathways at the microscale. We also study how this inter-scale feedback coupling alters algal biomass production and CO₂ sequestration at the reactor scale.

At the microscale, CO₂ enters the microalgal stromal cells in the form of HCO₃⁻, where the inorganic carbon is converted to organic carbon by RUBP in the presence of the RuBisCO enzyme to form 3-PGA (Huang et al., 2016), which undergoes a series of reactions to form G-3-P, and carbohydrates and lipids through gluconeogenesis and the TCA cycle, respectively. Simultaneously, acetate ions – the heterotrophic carbon source in our mixotrophic cultivation of *Chlorella sorokiniana* in 5L PBRs – enter the cytoplasm through the mct1 transporters and the glyoxylate cycle, producing TAGs (neutral lipids) through intermediates, such as malate, and a small amount of carbohydrates and proteins (Perez-Garcia et al., 2011). Thus, CO₂ sequestration from CO₂-rich flue gases and bioremediation of acetate-rich wastewater (Kargi and Arikian, 2013) can be simultaneously performed by the microalgae to produce bioenergy from lipids and carbohydrates and fertilizers from proteins.

The effectiveness of flashlight frequency in augmenting microalgal growth has been proven as a concept (Liao et al., 2014), which was

tested in various reactor designs, such as series lanterns shaped draft tubes (Ye et al., 2018), nested bottle 3D reactors (Kubar et al., 2023), and PBRs flash-lit (>10 Hz) by optical fibers inserted inside (Xue et al., 2013). However, reactors that use drafts and downcomers to boost flash frequencies have high pumping costs for driving the gas–liquid mixture up the draft and down the downcomer.

The bubble-column PBR design presented here is the first of its kind, which employs radial mixing to disperse the algal cells towards and away from the well-lit peripheral regions, thus simultaneously mitigating two major bottlenecks, namely, photo-limitation and photo-inhibition. Convective mixing in the radial direction in externally illuminated tubular PBRs mechanically engenders flashlight effect. The compressed air flow in the radial direction forces the algal cells to move back and forth between an illuminated concentric circle near the wall and a darker circle at the centre of the PBRs. This work hypothesizes that a radial mixing-mediated increase in the algal cell's shuttling frequency between the light zone and the dark zone not only enhances phototrophic synthesis but also synergizes heterotrophic carbon assimilation pathways by energizing the carrier molecules, such as NADPH and ATP. An earlier study hypothesized this inter-trophic synergy in algal photosynthesis but did not quantify it (Nair and Chakraborty, 2020). Here, we provide a method for quantifying inter-trophic synergy in mixotrophic microalgal growth, and show how this synergy occurring between the phototrophic and the heterotrophic pathways at the microscale could be augmented by manipulating the macro- or reactor-scale parameters, such as radial mixing, light intensity, and CO₂ and nitrogen loading. CFD simulations are performed for coupled turbulent flow in the gas spargers and laminar-bubbly flow of the culture medium in the PBR to quantify the flashlight frequency as a function of the inlet gas flow rate. This study offers a deployable optimization strategy for enhancing CO₂ sequestration in industrial-scale outdoor PBRs, especially those operating mixotrophically using acetate-rich wastewater bubbled with CO₂-rich flue gases.

2. Materials and methods

2.1. Materials

The culture of *Chlorella sorokiniana* was obtained from the Department of Biotechnology at our host institute. Magnesium Sulfate Heptahydrate, Calcium Chloride Dihydrate, Potassium Phosphate Dibasic, Potassium Dihydrogen Orthophosphate, Ammonium Molybdate Tetrahydrate, Cobalt Chloride Hexahydrate, Zinc Sulfate Heptahydrate, Ferrous Sulfate Heptahydrate, Boric Acid, Cobalt Chloride Hexahydrate, and Manganese Chloride Tetrahydrate were purchased from Loba Chemie Pvt. Ltd. Acetic Acid Glacial, Potassium Hydroxide Pellets, Copper (II) Sulfate Pentahydrate, and Ethylenedinitrilo Tetra-acetic Acid Disodium Salt (Dihydrate) were acquired from Merck Life Science Private Limited. Commercial urea and potash alum were procured locally. With the exception of urea and potash alum, all compounds were of analytical quality. The 5L bubble-column photobioreactor (PBR) was procured from Plastic Abhiyanta. The pH meter was acquired from Systronics Ltd.

2.2. Methods

2.2.1. Experimental Methods

The culture medium was prepared using urea as the organic nitrogen, which served as the external nitrogen source. In the preparation of the TAP media, we used urea as the nitrogen source rather than NH₄Cl. The concentration of the remaining solution, however, remained the same as in earlier published works (Ghosh et al., 2024). After 4 days of cultivation, a nitrogen-limited solution was used to refill the algal medium, wherein we cut off all the nitrogen sources (urea) during the preparation of the media. The initial concentration of acetic acid in the nutritional media was maintained at 1 mL/L (equivalent to 1.05 g/L). Since

microalgae thrive well in a basic environment with pH $\sim$ 7.5, the reactor pH was maintained as slightly basic using KOH solution. (Qiu et al., 2017). The solution was prepared using tap water in a 5L conical flask, which was autoclaved for 15 min at 121 °C to ensure total decontamination in an autoclave machine (Krishna Scientific Co.). The initial concentration of microalga in the culture medium was maintained between 0.11 and 0.24 g/L by combining an appropriate amount of inoculum with the cooled autoclaved media. The inoculum was stored in a 2L conical flask with the top tightly secured with cotton. The inoculum was replenished at regular intervals with enhanced nitrogen loading. The jars holding the media and the stainless steel spargers were also autoclaved during the preparation of the media. The 5L PBRs underwent sterilization using 99.9 % analytical-grade ethanol, followed by storage in a laminar hood exposed to UV light for nearly an hour to ensure complete de-contamination.

2.2.2. Photobioreactor (PBR) configurations with and without radial mixing

Two different types of PBR configurations with a reactor volume of 5L and a working volume of 4L are used here. The acrylic tubular bubble-column PBR of length 43.2 cm, has a diameter of 12.7 cm and a thickness of 5.1 mm. The externally illuminated PBR is housed within an aluminium jacket, with cool white fluorescent LED lights attached to its inner walls at a distance 7 cm from the reactor surface. In the first configuration, a bottom sparger is used to bubble the PBR from the bottom at a flow rate of 5 LPM. From here onwards, this configuration shall be referred to from as the control configuration, as can be seen in Fig. 1.a. In the second configuration, the total CO₂-enriched gas flow is redistributed into two parts, wherein the bottom sparger has a flow rate of 2 LPM. Concurrently, the radial sparger, as shown in Fig. 1.b., was facilitated with a flow rate of 3 LPM. The radial spargers made sure the microalgae cells could move back and forth between the well-lit edges of the PBR and the darker interior. This configuration (shown in Fig. 1.b) shall be referred to as a radially well-mixed configuration from now on. The CO₂ gas and sterilized air were passed separately through rotameters to achieve the desired CO₂ concentrations (%v/v) at the reactor inlet. The sterilization of the inlet CO₂-enriched air was achieved by passing the inlet gas and CO₂ through a PTFE membrane filter with a 0.2 mm pore size. The PBRs were lit externally using cool fluorescent LED lights.

2.2.3. Semi-batch cultivation process

Microalgal cultivation was performed in a semi-batch process for around 6 days (138 h). Prior studies show an illumination of around 10,500 lx to be optimal for microalgal growth in *Chlorella sorokiniana* (Mehta and Chakraborty, 2021). Hence, the same illumination was used for microalgal cultivation in this study using a diurnal cycle of 18:6 (L:D) since it yielded higher amount of lipids (Wahidin et al., 2013). However, at super-optimal light intensities that mimic the intensity of tropical sunlight, it has been found that tweaking the L:D ratio enhances the longevity of the algal biomass without inducing photoinhibition. Hence, for illuminations above 10,500 lx (i.e., 12,000 and 14,000 lx), the diurnal cycle was kept at a 12:12 L:D ratio to mitigate photoinhibition as long as possible. CO₂ inlet concentrations of atmospheric (0.04 %) CO₂ to 5 % CO₂ were used for microalgal cultivation at 10,500 lx. However, it was found that high illumination induced over-expression of CCM genes in the chloroplast even at moderate CO₂ concentrations (0.5–2 % CO₂ concentration) (Ruiz-sola et al., 2023). Since CCM is an energy intensive process (Reinfelder, 2011), CO₂ inlet concentrations ranging from optimal (3 %) to super-optimal (7 %) were employed in the study at super-optimal light intensities to make sure that diffusional transport, instead of CCM, mainly contributes to the whole inorganic carbon assimilation process. The supply of CO₂ to the PBR was cut-off from the air supply during the dark phase. The cultivation media was supplied with 2.5 mL/L of acetic acid-rich dosage every 3 h. During the dark phase of 6 h (10,500 lx) and 12 h (12,000 and 14,000 lx), a proportional

amount of acetic acid-rich dosage was supplied to the culture media just before the onset of the dark phase. All the experiments were performed at an ambient temperature of 25 °C. The semi-batch cultivation processes, analysis of dry cell weight data and the their macromolecular and acetic acid analysis were performed in duplicates to ensure consistent results ($n = 2$), and analysed using one-way ANOVA.

2.2.4. Determination of dry cell weight and microalgal growth

The biomass concentration, or the dry cell weight, was determined optically by measuring the absorbance of the algal cells using a UV-vis spectrophotometer (UV-1900, UV Spectro-photometer, Shimadzu, Japan) at a wavelength of 682 nm (Kumar and Das, 2012). The absorbance of known concentrations of freeze dried algae in deionized water was measured. Subsequently, the absorbance values (x-axis) obtained were plotted against different known biomass concentrations (y-axis), resulting in the generation of a calibration curve. This process was repeated three times, and the average value of absorbance was taken for generation of the calibration curve. The biomass concentration was successively determined by applying the equation: Dry cell weight (g/L) = $0.4425 \times \text{Absorbance}_{682\text{nm}}$ ($R^2 = 0.9934$), in cases where direct measurement of dry cell weight was not possible.

2.2.5. Determination of macromolecular content and composition

The macromolecular composition was determined using the procedure described below.

2.2.5.1. Lipid estimation. Dried algal biomass (50 mg) at varied CO₂ concentrations was mixed with 2 mL of *n*-Hexane: Isopropanol (3:2 v/v) and extracted at 140 rpm for 7 h on a shaker (Model: RS-24BL, Remi, India). After centrifugation at $6705 \times g$ for 5 min at 25 °C, the supernatant was obtained, and shaken for another 7 h at 140 rpm using 2 mL of *n*-Hexane: Isopropanol (2:3 v/v) solution, after which the supernatant from the second extraction cycle was combined with the first. The supernatant solution was then combined with an equal volume of water: 5 N NaCl (9:1). The solutions were vortexed for 15–20 min to separate the neutral lipid-containing hexane layer from the top. The lipid content was calculated by drying the hexane layer below 40 °C and performing gravimetric measurements (Hara and Radin, 1978).

2.2.5.2. Protein estimation. Protein content was determined using modified Lowry's method. 5 mg dried algal biomass was mixed with 1 N NaOH. The solution was cooled after 10 min at 100 °C, and 2.5 mL of a solution of 50 mL of 5 % Na₂CO₃ and 2 mL of 0.5 % CuSO₄ in 1 % Na-K Tartarate were added. The solution was incubated and vortexed for 15 min with 0.5 mL of 1 (N) Folin's reagent. The final protein content was measured by comparing the solution's absorbance at 650 nm with a previously-generated Bovine Serum Albumin calibration curve (Lowry et al., 1951).

2.2.5.3. Carbohydrate estimation. Carbohydrate content was measured using phenol-sulphuric acid method. 1 mg/mL microalgal sample was mixed with 5 % (v/v) phenol, and 3 mL H₂SO₄ was added. After heating the solution for 5 min at 90–100 °C, its absorbance was measured at 490 nm. The total carbohydrate content was calculated using a known pure dextrose calibration curve (Masuko et al., 2005).

2.2.6. Determination of acetic acid concentration in the bulk media

The microalgal samples collected at regular intervals were centrifuged at $6705 \times g$ at 25 °C for 7 min. Thereafter, the absorbance of the supernatant was analysed in a UV-vis spectrophotometer (UV – 1900, UV Spectrophotometer, Shimadzu, Japan) at 204 nm. Initially, various known acetic acid concentrations were calibrated against absorbance at the same wavelength. The final acetic acid concentration left in the media was obtained using the relationship: $[\text{CH}_3\text{COOH}] \text{ (mL/L)} = 4.706 \times \text{Absorbance}_{204\text{ nm}} + 0.0647$ ($R^2 = 0.9983$).

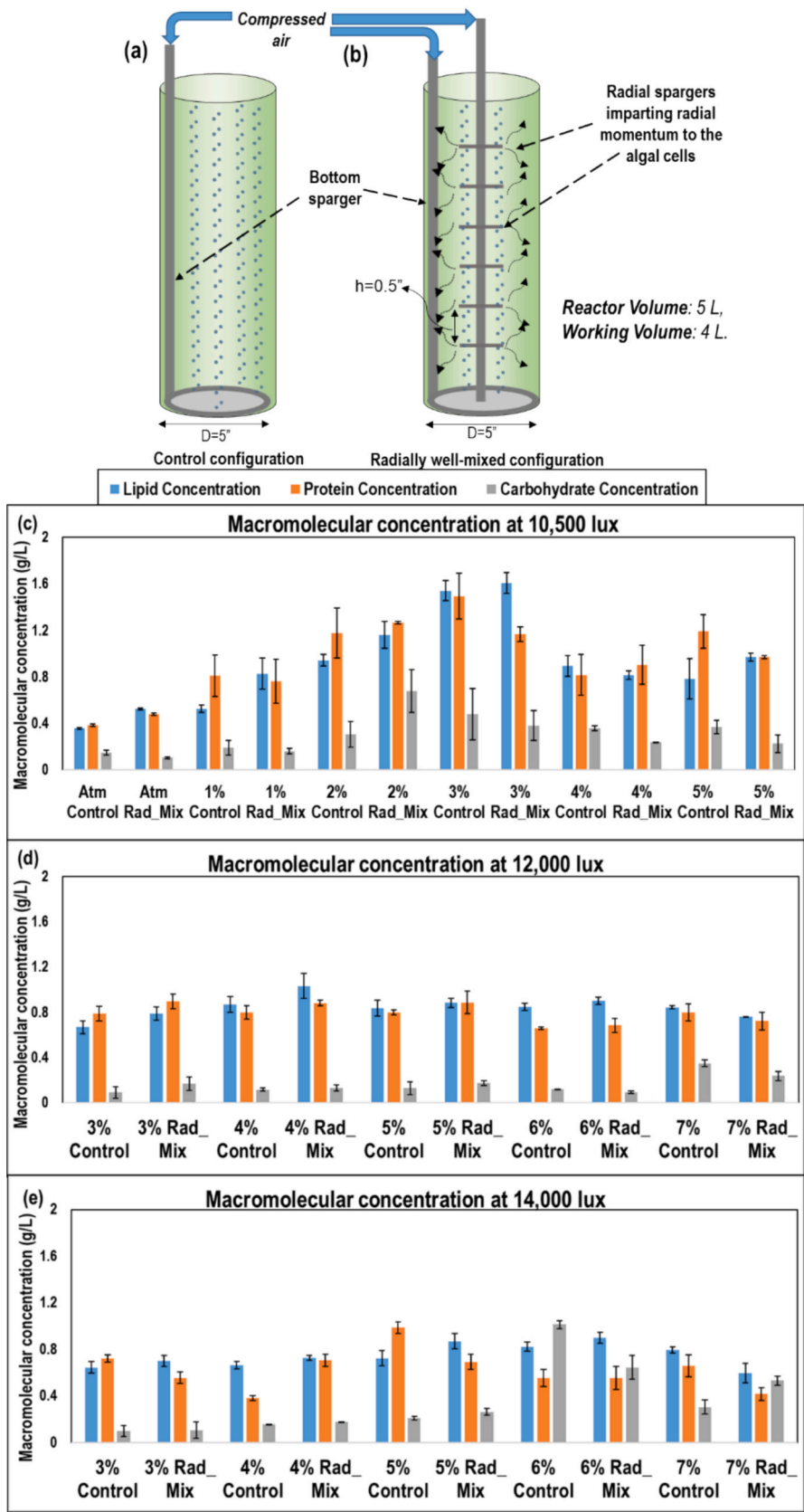


Fig. 1. Reactor configurations used: (a) Control reactor and (b) Radially-mixed reactor, and macromolecular concentration (post day 6) at various CO₂ concentrations (c) 10,500 lx, (d) 12,000 lx, (e) 14,000 lx [n = 2; please see supplementary material for a statistical analysis of the data].

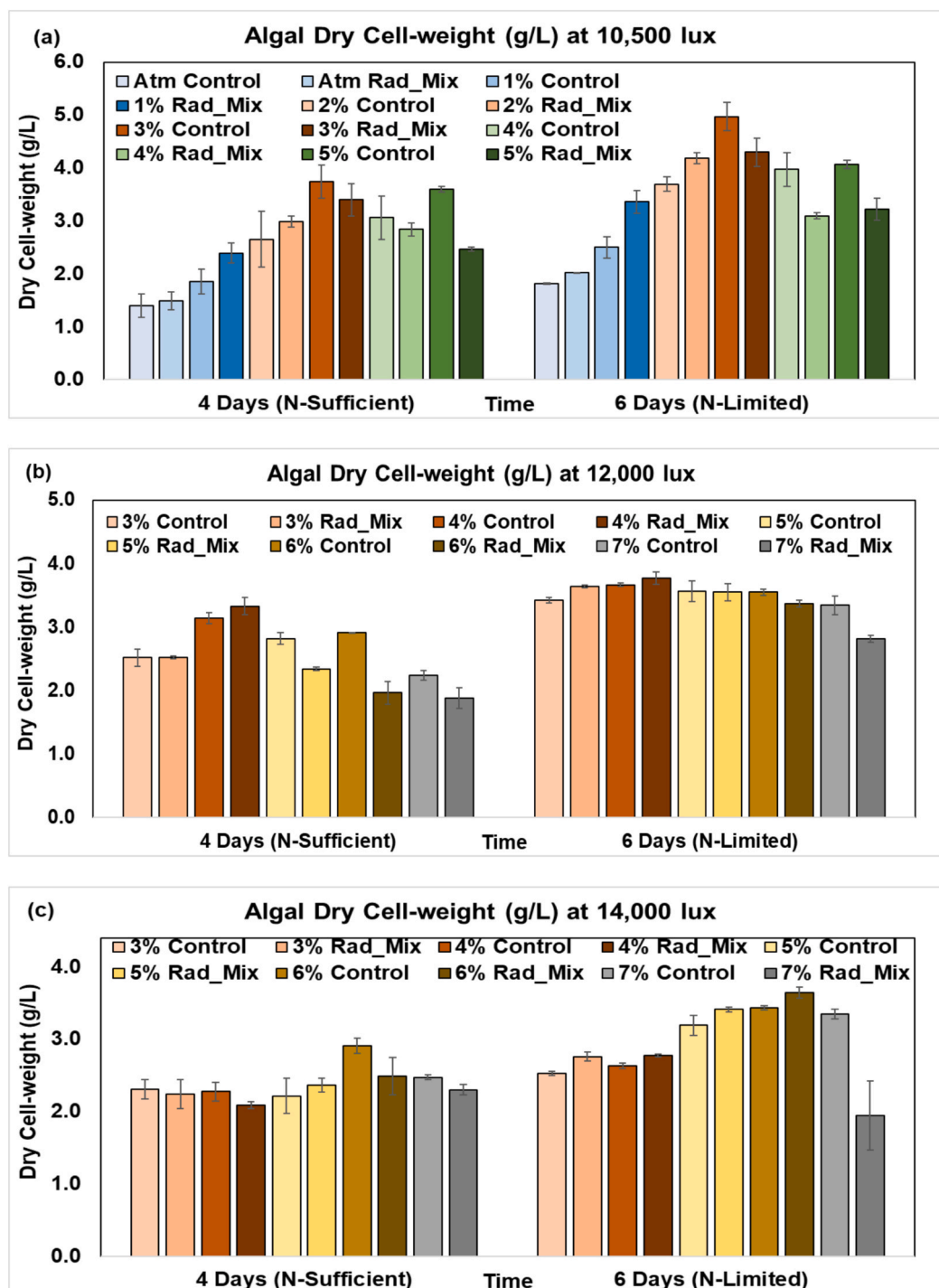


Fig. 2. Dry microalgal biomass weight obtained at various CO₂ concentrations at (a) 10,500 lx, (b) 12,000 lx, (c) 14,000 lx [n = 2; please see supplementary material for a statistical analysis of the data].

2.2.7. Simulation Methods

The fluid flow pattern in the bubble column PBR was simulated using COMSOL Multiphysics 5.4 software, by considering turbulent flow of gas in the spargers and laminar-bubbly flow of liquid in the PBR. Reynolds-averaged Navier-Stokes (RANS) equation was used to model momentum conservation in turbulent flow in the radial and axial spargers and determine the outlet velocities in the radial spargers at various heights. The RANS equation is closed by employing the k - ϵ model to estimate the turbulence kinetic energy (k) and the rate of dissipation (ϵ). The mean flow characteristics in the radial spargers are simulated by solving the closed RANS equation with suitable initial and boundary conditions,

such as varying inlet velocities (air flowrates) at the sparger entrance and no-slip condition at the sparger walls. The interfacial tension between the two phases was also accounted for in the RANS equation. Subsequent to determining the outflow velocities from the spargers at various heights, the liquid flow profile inside the PBR was simulated using the Navier-Stokes equation in the laminar bubbly-flow regime, with the no-slip boundary condition at the reactor walls. Thus, the turbulent gas flow profile in the spargers and the liquid flow profile in the laminar-bubbly regime in the PBR were simulated at various air flow rates ranging from 4 to 7 LPM for both the radially-mixed and the control reactor configurations.

2.2.8. Determination of elemental carbon content

The carbon content (%) of the dried algal biomass was measured using Dumas method (da Silva et al., 2024) in a Vario macro cube CHNS analyser. The biomass was passed through a series of containers that sequentially removed the carbon, nitrogen, sulfur and hydrogen components. The elemental fractions were multiplied with the mass of the dried biomass to obtain the elemental content of the algal biomass.

3. Results and discussion

First, we investigate how changes in reactor-scale parameters, namely radial mixing, organic (acetate) and inorganic (CO_2) carbon loading, and external illumination, alter the outputs at the *meso* (macromolecular) and micro (elemental) scales and affect inter-trophic synergy. Then we present a method for quantifying synergy between phototrophic CO_2 sequestration and heterotrophic carbon assimilation, using which we identify the macroscale parameters capable of enhancing inter-trophic synergy and CO_2 sequestration.

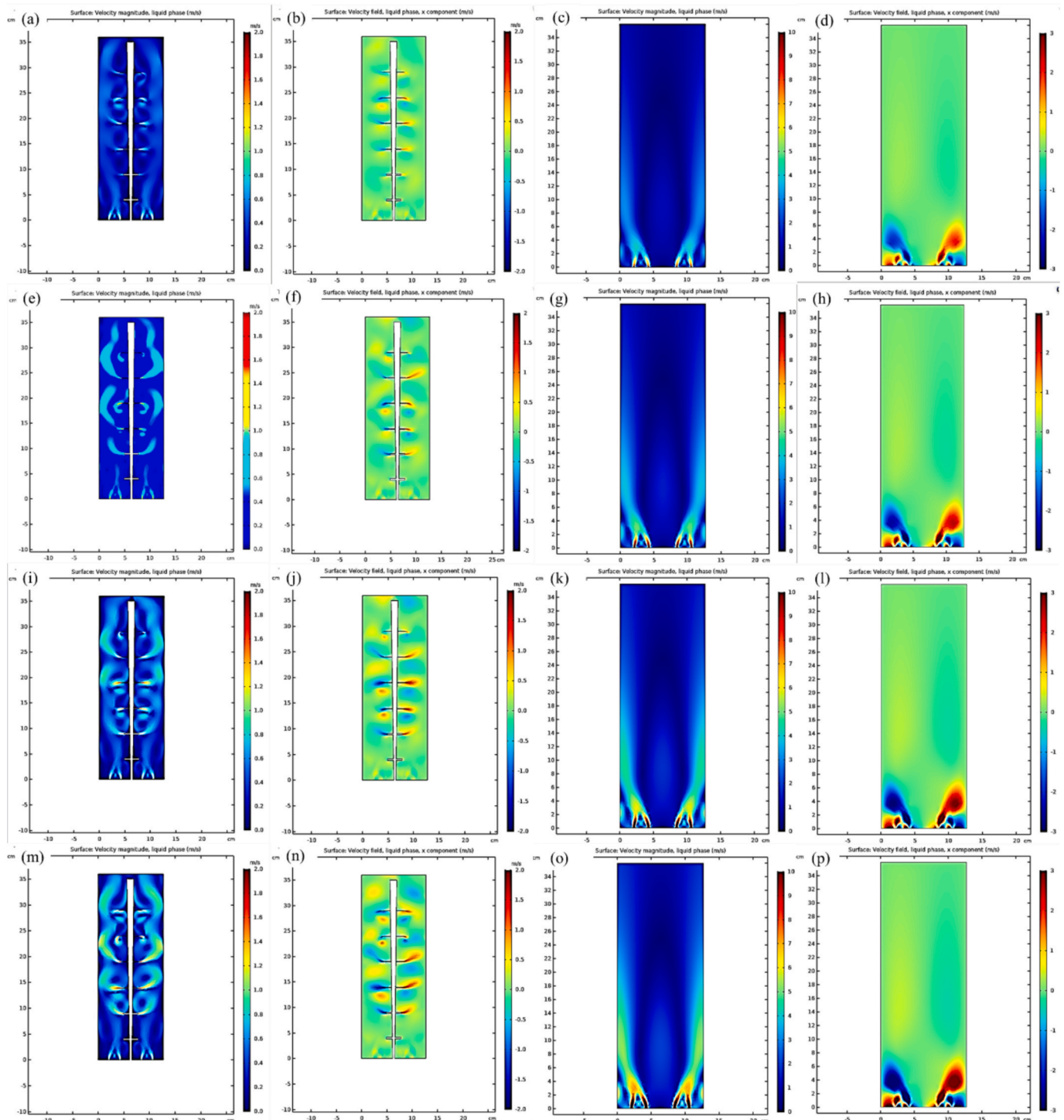


Fig. 3. Velocity magnitude and its radial component inside the photobioreactor for radially-mixed (a-b, e-f, i-j, m-n) and control (c-d, g-h, k-l, o-p) configurations at inlet gas flowrates of: (a-d) 4 LPM, (e-h) 5 LPM, (i-l) 6 LPM, (m-p) 7 LPM (please zoom in for details).

3.1. Effect of radial mixing across the scales

Radial mixing ensures that the CO₂-rich compressed air moves the microalgal cells and the nutrient-rich media back and forth across the reactor radii. Imparting radial momentum allows the microalgal cells as move against the light intensity gradient from the darker internal areas to the well-illuminated peripheral areas.

3.1.1. Effect of radial mixing on liquid flow velocity in the PBR

The impact of radial mixing on transverse circulation of liquid in the reactor is illustrated by comparing the velocity profiles (including the magnitude of velocity and its radial component in Fig. 3) of the control and radially well-mixed configurations within the PBR. As the inlet flow rate increases, the average radial velocity (v_{ml}) of the liquid phase in a radially-well mixed configuration increases from almost 0.9 m/s at 4 LPM to 1.5 m/s at 5 LPM, 1.7 m/s at 6 LPM, and 1.92 m/s at 7 LPM. These values are 3–5 fold higher than the v_{ml} in the control configuration, wherein the respective v_{ml} values are 0.17 m/s (at 4 LPM), 0.33 m/s (at 5 LPM), 0.50 m/s (at 6 LPM) and 0.66 m/s (at 7 LPM).

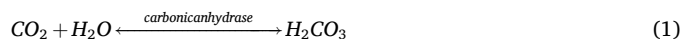
After simulating the average liquid phase radial velocities (v_{ml}) along the height of the PBR at various air flowrates, the distance (s) between the sparger end (radial sparger outlet) and the reactor wall is determined. Assuming no-slip condition between the algal cells and the liquid, the time (t_f) taken for one roundtrip by the algal cells is calculated as $t_f = 2 \times s/v_{ml}$, and the inverse of t_f gives the flash frequency (f_f) as $f_f = 0.5v_{ml}/s$. Consequently, increasing the inlet flow rate in a radially-mixed reactor increases the average flash frequency from 7.67 Hz (at 4 LPM) to 12.5 Hz (at 5 LPM), 14 Hz (at 6 LPM), and 16 Hz (at 7 LPM), while a similar increase in the control case increases the average flash frequency from 1.4 Hz (at 4 LPM) to 2.75 Hz (at 5 LPM), 4.17 (at 6 LPM) and 5.5 Hz (at 7 LPM). These simulated values of the algal flashlight frequency demonstrate that introducing radial mixing in the PBR increases the flashlight effect by three to five folds, allowing the algal cells to shuttle faster between the light and the dark zones. While the radial fluid velocity directs the cells from the dark zone to the light zone, the peripheral walls of the reactor redirects their momentum back towards the centre. This flashlight effect increases microalgal productivity, if the liquid pH remains within an optimal range (between 7.5–9). On the other hand, the control reactor restricts the radial circulation merely to the reactor bottom (Fig. 3.d,h,l,p).

It may be theoretically possible to mimic the radial velocity distribution achieved at low to moderate air flowrates in a radially-mixed reactor by drastically increasing the air flowrate in a control reactor. However, this would produce high shear stress that can degenerate the algal cells (Saini et al., 2018). Hence, we optimize two rheological parameters, namely, radial velocity and shear stress, to operate the reactor at an inlet velocity that minimizes cell damage while maintaining high flashlight frequencies. We find that super-optimal (> 6 LPM) gas-flow rates that produce high mean radial velocities (>1.7 m/s) lead to the onset of the death phase on 4th (or the 5th) day, which is unintended, since lipid production begins on the 4th day (Bekirogullari et al., 2018). Thus, the experiments performed in this study are conducted at an inlet gas flowrate of 5LPM, which provides sufficient radial mixing, a high mean radial velocity of 1.5 m/s and an average flash frequency of 12.5 Hz.

3.1.2. Effect of radial mixing on the extracellular pH

The saturation concentration of dissolved CO₂ in water at 25 °C is 1.449 g/L, while the equilibrium concentrations of dissolved CO₂ using Henry's Law at 0.04 % (atmospheric), 4 %, 5 %, 6 %, and 7 % CO₂ in the gas phase are 0.0006 g/L, 0.0596, 0.0745 g/L, 0.0894 g/L, and 0.1043 g/L, respectively. Therefore, radial mixing homogenizes and increases the dissolved CO₂ concentration across the reactor cross-section, leading to a significant reduction in the extracellular pH (due to the production of carbonate, bi-carbonate and H⁺ ions (Iliuta et al., 2023)) as compared to the control, regardless of the external

illumination and the CO₂ loading in the reactor.



The pH increases considerably during the dark phase of 6 h (at 10,500 lx) and 12 h (at 12,000 and 14,000 lx) when the reactor runs at atmospheric CO₂, leading to a significant increase in the average pH over the entire span of 138 h in the semi-batch cultivation process (see supplementary material). High extracellular pH acidifies the intracellular cytoplasmic space as the protons enter the stromal space and interferes with algal metabolism, reducing biomass growth at extracellular pH < 7.5 (Mehta and Chakraborty, 2021; Van Amerongen and Croce, 2013).

Above a critical inlet CO₂ loading, the radially-mixed reactor has average extracellular pH values much lower than the control for a considerable span of the semi-batch cultivation process, leading to CO₂ inhibition of algal growth. Noticeably, this critical inlet CO₂ loading increases with increasing reactor illumination: 2 % at 10,500 lx, 4 % for 12,000 lx, 6 % for 14,000 lx. Moreover, at any inlet CO₂ loading, higher illumination increases the reactor pH; e.g., at 3 % CO₂, the time-averaged pH in a radially mixed reactor increases from 7.57 at 10,500 lx to 7.81 at 12,000 lx, and 7.86 at 14,000 lx. This is consistent with earlier findings (Venn et al., 2019) that well-illuminated algal solutions have higher pH than those exposed to lower light intensities. Higher uptake of CO₂ with a simultaneous increase in the extracellular pH at higher illumination points to light being a limiting substrate in phototrophic algal growth, especially in the dark zones at the reactor core. Thus, a tetrad of engineering design strategies – higher illumination and nitrogen limitation with higher CO₂ input in radially-mixed reactors that raise light penetration and CO₂ transport across the reactor cross-section – ensures that neither of the two substrates would limit phototrophic biomass growth.

3.1.3. Effect of radial mixing and illumination on macromolecular composition

While the algal biomass concentration peaks at a dynamic pH of 7.5, the lipid content in *Chlorella sorokiniana* peaks at a pH of 7–7.5 (Qiu et al., 2017). Thus, at a pH > 7.5, radial mixing enhances the overall lipid amount. Fig. 1.c, d, and e show the lipid content at 10,500 lx peaks at 3 % (radially well mixed case: 1.61 g/L, control: 1.54 g/L) at 10,500 lx. The lipid contents at 12,000 lx and 14,000 lx peak at 4 % (radial well-mixed case: 1.03 g/L, control: 0.87 g/L) and 6 % CO₂ inlet concentrations (radially well-mixed case: 0.82 g/L, control: 0.90 g/L), respectively. Moreover, enhanced illumination facilitates heterotrophic assimilation of organic carbon from acetic acid through the glyoxylate pathway (Ghosh et al., 2024). Carbohydrate and protein contents are weakly influenced by reactor scale parameters such as CO₂ loading, reactor illumination and mixing (Fig. 1.c,d, and e), and they follow the trend of the biomass growth with increasing CO₂ and light.

3.1.4. Effect of radial mixing and illumination on elemental carbon concentrations in algal cells

In cases where the pH and the CO₂ concentrations lie in the favourable zone, radial mixing increases the carbon content and the microalgal biomass growth by synergizing both autotrophic (see supplementary material) and heterotrophic pathways at the molecular (micro) scale.

Since the timescale for the light-dependent reactions is a fraction of a second (Belyaeva et al., 2011; Van Amerongen and Croce, 2013), the PS II and PS I centres in the thylakoid membrane require momentary exposure to high intensity light, leading to higher phototrophic growth. Radial mixing provides the algal cells with enough time to recuperate

from exposure to super-optimal light intensity, helping them regenerate the damaged PS II and PS I centres in centi-seconds (Abu-Ghosh et al., 2016). The regenerated PS II and PS I centres fuel the uninterrupted carbon assimilation through the Calvin cycle (Nair and Chakraborty, 2020), the glyoxylate cycle (Geng et al., 2023), the GS/GOGAT cycle (Fathy et al., 2023; Lea and Mifflin, 2003), and gluconeogenesis (Venkata Subhash et al., 2017) in the light-independent phase. Thus, radial mixing in higher illumination produces higher autotrophic and total carbon content (Fig. 6 and supplementary materials) by reducing photo-inhibition and circumnavigating self-shading among the algal cells. As in the case of total biomass output, the maximum carbon content is obtained at 3 % CO₂ concentration (control: 2.15 g/L, radial mixing: 1.83 g/L). The carbon content generally increases in the radially-mixed case until 2 % inlet CO₂ concentration, above which the control configuration shows higher carbon content due to lower CO₂ inhibition.

At 12,000 lx, the radially-mixed reactor assimilates more carbon (till

4 %, post which carbon-dioxide inhibition occurs), with the following trend: 4 % CO₂ (control: 1.46 g/L, radially-mixing: 1.62 g/L) > 3 % CO₂ (control: 1.11 g/L, radially-mixing: 1.54 g/L) > 5 % CO₂ (control: 1.42 g/L, radially-mixed: 1.42 g/L) > 6 % CO₂ (control: 1.40 g/L, radially-mixed: 1.32 g/L) > 7 % CO₂ (control: 1.27 g/L, radially-mixed: 1.16 g/L). At 14,000 lx, the radially well-mixed configuration always shows higher carbon assimilation, except for 7 % CO₂, when the advantage of reduced photo-damage of the algal cells imparted by radial mixing is outweighed by the increased acidification by very high CO₂ loading. The final elemental carbon concentrations obtained after 6 days of cultivation at 14,000 lx follows the trend: 6 % CO₂ (control: 1.27 g/L, radially-mixed: 1.56 g/L) > 7 % CO₂ (control: 1.23 g/L, radially-mixed: 0.68 g/L) > 5 % CO₂ (control: 1.15 g/L, radially-mixed: 1.38 g/L) > 4 % CO₂ (control: 0.86 g/L, radially-mixed: 1.03 g/L) > 3 % CO₂ (control: 0.76 g/L, radially-mixed: 0.92 g/L).

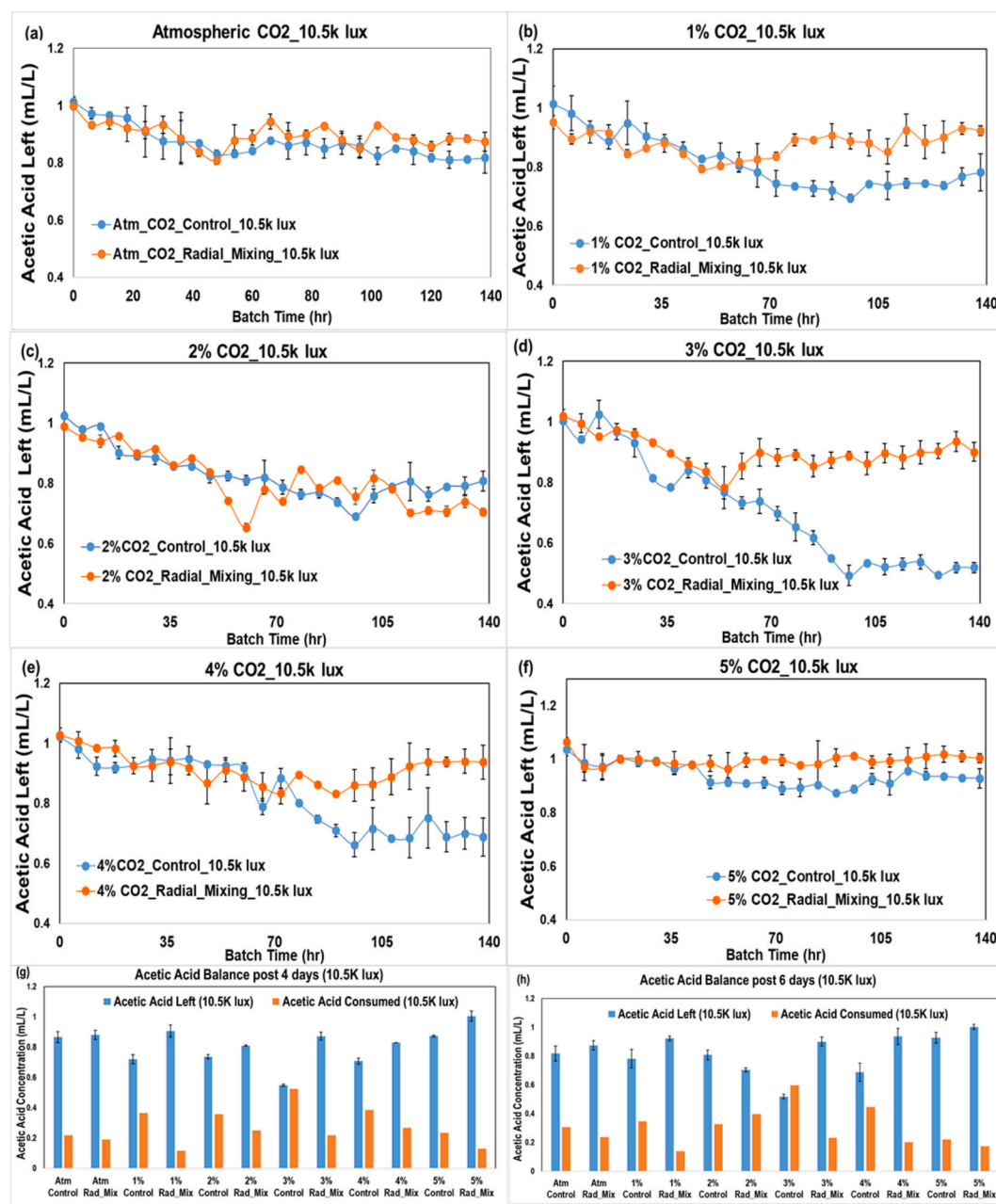


Fig. 4. Acetic acid dynamics at 10,500 lx at inlet CO₂ percentages of (a) 0.04 %, (b) 1 %, (c) 2 %, (d) 3 %, (e) 4 %, (f) 5 %. Acetic acid remaining and consumed after: (g) 4 days of N-abundance, (h) 6 days of N-limited at 10,500 lx. (Fig. 4.a-f. show initial 96 h of N-abundance, followed by 42 h of N-limitation) [n = 2].

3.2. Effect of acetic acid consumption on algal growth

Acetic acid – the organic carbon source used in this study – enters the cytoplasm as acetate ions, transported by the bidirectional mct1 transporters (Perez-Garcia et al., 2011), which can also extrude acetate from the cytoplasm when its concentration exceeds the threshold required to maintain a favourable intracellular pH (Liu et al., 2016). The initial concentration of acetic acid in the extracellular medium is maintained at 1 mL/L. Subsequently, this concentration varies depending on the microalgal uptake and the three-hourly addition of acetic acid. Excess acetic acid hampers marine ecological systems (Jasni et al., 2020). Thus, the microalgal uptake of acetic acid and its assimilation into lipids through the glyoxylate cycle (Ahn et al., 2016) facilitate both water security and biodiesel production.

3.2.1. Acetic acid dynamics and consumption at 10,500 lx

Measuring the depletion rate of acetic acid from the extracellular medium allows us to quantify heterotrophic growth in the mixotrophic cultivation. The photosynthetic light reactions not only fuel inorganic

carbon assimilation in the Calvin cycle but also energize the heterotrophic carbon assimilation pathways by supplying ATP and NADPH.

Fig. 4 a–f show how the acetic acid dynamics vary in the extracellular medium at 10,500 lx, while Fig. 4, g and h show the acetic acid balance at the end of the 138 h of mixotrophic cultivation. At 10,500 lx, only the autotrophic carbon assimilation pathway is amplified by radial mixing until day 4 in cases where extra-cellular pH > 7.5. The lipid synthesis from acetic acid is generally energized only at higher illuminations (He et al., 2015). Regardless of the CO₂ loading in the reactor, acetic acid consumption by the algae is consistently higher in the control reactors than the radially-mixed ones (Fig. 4, g–h), suggesting the strong preference that radial mixing has in amplifying phototrophy (see supplementary material) over heterotrophy at the optimal light intensity of 10,500 lx. The acetic acid uptake peaks at 3 % CO₂ (radial mixing: 0.6 mL/L, control: 0.23 mL/L), closely followed by that at 2 % CO₂ (radial mixing: 0.33 mL/L, control: 0.40 mL/L).

3.2.2. Acetic acid dynamics and consumption at 12,000 lx

Fig. 5 shows that the maximum acetic acid consumption of 0.54 mL/

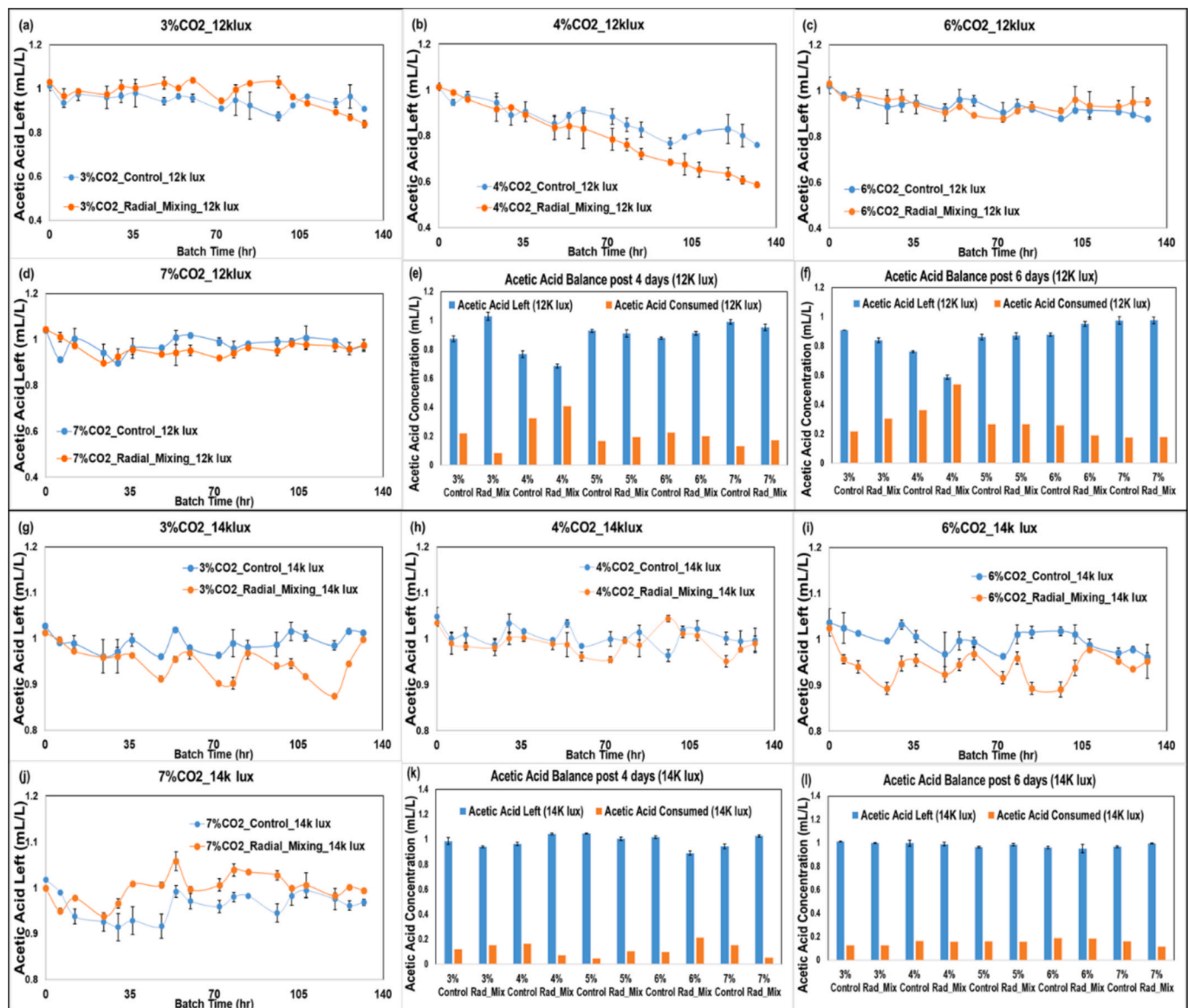


Fig. 5. Acetic acid dynamics at 12,000 and 14,000 lx and CO₂ inlet of (a,g) 3%, (b,h) 4%, (c,i) 6%, (d,j) 7%. Acetic acid left and consumed after: 4 days of N-abundance at (e,k) 12,000 and 14,000 lx, and 6 days of N-limitation at (f,l) 12,000 and 14,000 lx. (Fig. 5, a–d. and 5. g–j. show initial 96 h of N-abundance, followed by 36 h of N-limitation) [n = 2].

L (Fig. 5.f) and the growth algal biomass concentration (Fig. 2.b) both correspond to 4 % CO₂ in the radially-mixed reactor after 6 days of mixotrophic cultivation. Super-optimal illumination (12,000–14,000 lx) promotes synergy between phototrophic and heterotrophic growth. The pathway for assimilating acetic acid into TAGs (neutral lipids) is significantly more active at higher light intensities (He et al., 2015). At 12,000 lx and 3 % and 4 % CO₂ loading, radial mixing produces higher microalgal biomass output at the macroscale (Fig. 2.b) and higher lipid output at the mesoscale (Fig. 1.d), suggesting improved synergy between phototrophy and heterotrophy, resulting in enhanced CO₂ sequestration and acetate uptake, respectively.

3.2.3. Acetic acid dynamics and consumption at 14,000 lx

At 14,000 lx, the acetic acid consumption decreases to a maximum uptake value of 0.18 mL/L in the case of 6 % CO₂ concentration (control case) at the end of day 6 of mixotrophic cultivation (Fig. 5.l). The acetic acid consumption in the radially-mixed reactor at the end of day 6 is generally found to be lower than its counterpart (Fig. 5.g-j). However, the radially-mixed reactors with various CO₂ loading is found to be better at acetate uptake than their (control) counterparts (Fig. 5.g-l) until 4 days, suggesting that radial mixing promotes inter-trophic synergy. Higher lipid content in radially-mixed reactors (Fig. 1.e) further corroborates that hypothesis. The acetate uptake at the end of day 4 and day 6 peak at 6 % CO₂ in radially-mixed (uptake amount: 0.21 mL/L), and 6 % CO₂ in control (uptake amount: 0.19 mL/L) reactors, respectively.

3.3. Effect of CO₂ concentration on algal growth

Atmospheric CO₂ present enters the chloroplast in the form of HCO₃⁻ through an active transport process (Huang et al., 2017; Witte, 2011). At concentrations above 1 %, such as those in industrial flue gases requiring

for 10,500 lx, 12,000 lx, and 14,000 lx, respectively. Thus, for all the illuminations, excess CO₂ inhibits the algal metabolism by interfering with the intra-cellular pH (Moheimani, 2013).

Fig. 2 shows how super-optimal illumination and super-optimal CO₂ loading in the reactor affects the biomass output. Radial mixing homogenizes the H⁺ and CO₂ concentrations across the reactor scale more than in the control cases. As soon as the dynamic extracellular pH falls below 7.5, an imbalance between the extra- and the intra-cellular pH compels the algal cells to block CO₂ influx to their intra-cellular space by modifying the cell membrane into proton barriers (Choi et al., 2021). This produces two kinds of regimes for all light intensities: (a) radial mixing facilitates algal proliferation at reactor pH > 7.5, and (b) radial mixing inhibits algal proliferation more compared to its control counterpart at reactor pH < 7.5. In the following section, we shall show how a tetrad of operating parameters – external illumination, CO₂ loading, nitrogen limitation and radial mixing – may be manipulated to promote inorganic carbon sequestration and inter-trophic synergy in the microalgae.

3.4. Quantifying inter-trophic synergy

3.4.1. Stratifying total elemental carbon assimilated into its trophic components

The phototrophic and heterotrophic components in the total mixotrophic biomass growth may be quantified by estimating the acetic acid consumption by the algae (Mehta and Chakraborty, 2022). The acetic acid consumed at the end of the six days of mixotrophic cultivation is shown in Figs. 4, and 5. Thereafter, eqns. (4) – (6) are used to separate the overall elemental carbon assimilated through its multi-trophic modes into its phototrophic (inorganic) and heterotrophic (organic) components and plot them in Fig. 6.a-i.

$$CH_3COOH_{consumed} \left(\frac{g}{L} \right) = \left\{ CH_3COOH_{supplied} \left(\frac{mL}{L} \right) - CH_3COOH_{left} \left(\frac{mL}{L} \right) \right\} \times 1.05 \left(\frac{g}{mL} \right) \quad (4)$$

sequestration, CO₂ enters the stromal space primarily through the diffusional transport (Zhao et al., 2015). Thereafter, CO₂ reacts with RUBP at the site of the RuBisCO enzyme in the chloroplast to form 3-PGA. The 3-PGA molecule then undergoes a series of transformations, leading to the production of glyceraldehyde-3-phosphate (G-3-P). This G-3-P molecule then enters the cytoplasm and forms carbohydrate and lipids through gluconeogenesis and TCA cycles, respectively (Ghosh

where the density of CH₃COOH is 1.05 ($\frac{g}{mL}$).

$$OrganicC_{consumed from CH_3COOH} \left(\frac{g}{L} \right) = \left(\frac{24}{60} \right) \times CH_3COOH_{consumed} \left(\frac{g}{L} \right), \quad (5)$$

$$InorganicC_{sequestered from CO_2} \left(\frac{g}{L} \right) = TotalC_{Content} \left(\frac{g}{L} \right) - OrganicC_{consumed from CH_3COOH} \left(\frac{g}{L} \right). \quad (6)$$

et al., 2024). This results in the overall growth of the microalgal cells. Thus, higher phototrophic assimilation of CO₂ is expected to increase the overall biomass concentration (Fig. 2). As mentioned in Section 3.1.1, the gas phase concentrations of CO₂ (0.04–7 %) used in this study are far below the saturation gas-phase concentration of CO₂ at 25 °C corresponding to the maximum concentration (1.449 g/L) of dissolved CO₂ in water. This implies that higher input gas-phase CO₂ loading corresponds to higher dissolved CO₂ concentration in the reactor. However, as shown in Section 3.1.1, high dissolved CO₂ concentration acidifies the reactor medium and lowers the extracellular pH below 7.5, slowing down algal proliferation. This happens at 3 %, 4 % and 5 % CO₂

As may be observed from Fig. 6.a-i, heterotrophic growth contributes to a small fraction of the overall growth. The maximum heterotrophic growth is observed at 10,500 lx illumination, in which we also obtain the maximum overall biomass concentration (Fig. 2). The heterotrophic growth contribute about to a minimum of 4 % (for 5 % CO₂) to a maximum of 26 % (for 1 % CO₂) of the total mixotrophic growth. Both the phototrophic and heterotrophic carbon assimilation (in g/L) peak at 3 % CO₂ and 10,500 lx for the control case (Fig. 6), with values of 1.85 and 0.24 g/L, respectively, after 6 days of mixotrophic cultivation. The radially-mixed reactor, on the other hand, shows lower assimilation of inorganic (1.77 g/L) and organic (0.08 g/L) carbon content, due to lower

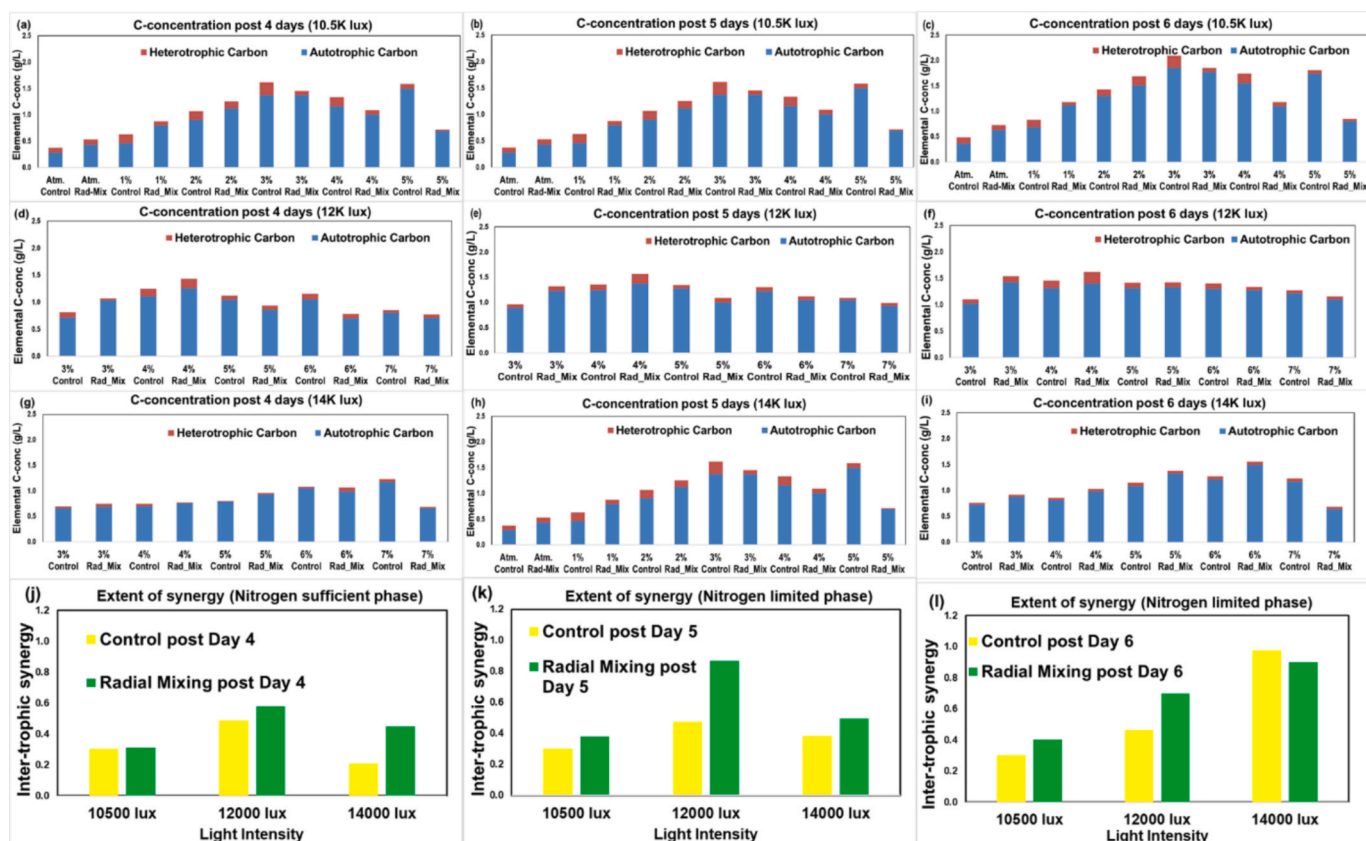


Fig. 6. Contribution of autotrophy and heterotrophy in total elemental C-content post Day 4, 5 and 6 at various inlet CO₂ % at (a,b,c) 10,500 lx, (d,e,f) 12,000 lx, (g, h,i) 14,000 lx; and quantification of inter-trophic synergy by Pearson correlation coefficient post (j) Day 4 (with N-sufficiency), (k) 5 days (with N-limitation) and (l) 6 days (with N-limitation) (please zoom in for details).

extra-cellular pH. This is closely followed by biomass obtained at 2 % CO₂, where the inorganic carbon concentrations are 0.68 and 1.12 g/L for the control and the radially-mixed configurations, respectively, and the organic carbon assimilated are 0.15 and 0.06 g/L for the control and the radially-mixed configurations, respectively. At the optimal illumination of 10,500 lx, we observe that radial mixing largely enhances the phototrophic carbon assimilation pathways.

At super-optimal illumination, the acetic acid uptake, like its overall biomass growth, peaks at 4 % CO₂ (radially-mixed case) and 6 % CO₂ (radially-mixed case) concentrations at 12,000 lx and 14,000 lx, respectively (Fig. 5). The enhanced illumination levels not only influence the phototrophic pathways but also improves the heterotrophic carbon uptake pathway through glyoxylate cycle. At 12,000 lx and 4 % CO₂ loading, the inorganic carbon assimilated reduce to 1.31 and 1.40 g/L for the control and the radially-mixed reactors, respectively, while the organic carbon assimilated increase to 0.15 and 0.22 g/L for the control and the radially-mixed configurations, respectively, suggesting improved synergy between phototrophy and heterotrophy. On the other hand, at 14,000 lx, the carbon assimilation peaks at 6 % CO₂ concentrations, with inorganic carbon uptake of 1.21 and 1.49 g/L for the control and the radially-mixed configurations, respectively, and organic carbon uptake of 0.060 and 0.064 g/L for the control and the radially-mixed reactors, respectively.

Though the overall phototrophic CO₂ sequestration is higher at 10,500 lx, super-optimal light, facilitated by lower operational costs (since they operate 6 h lesser than the 10,500 lx case), enables comparable CO₂ assimilation (Fig. 6.d, 6.f). We infer that while radial mixing facilitates inorganic carbon sequestration alone at optimal levels (10,500 lx), it promotes both phototrophic and heterotrophic carbon assimilation at higher illumination (12,000–14,000 lx). Moreover, super-optimal illumination with a 12:12 light: dark cycle mimics the

algal circadian rhythm at par with sunlight (Viruela et al., 2016), thereby increasing the energy efficiency over the 18:6 light: dark cycle suited for optimal illumination, and making this process suitable for outdoor cultivation.

We use an effectiveness factor ($\eta = \text{CO}_2 \text{ sequestered with radial mixing} / \text{CO}_2 \text{ sequestered without radial mixing}$) to quantify the effectiveness of radial mixing in sequestering CO₂ at optimal to super-optimal light. Radial mixing helps sequester 72 % more CO₂ at atmospheric CO₂ and 10,500 lx (18:6 light: dark), 40 % more at 12,000 lx and 3 % CO₂, and 23 % more at 14,000 lx and 6 % CO₂ for 12:12 circadian cycles (see supplementary material).

3.4.2. Quantification of inter-trophic synergy at the reactor-scale

The multiscale synergy between the autotrophic and the heterotrophic pathways starts with ATP and NADPH transport in the chloroplast scales up through the metabolic reactions for macromolecular synthesis at the mesoscale and the algal biomass concentration at the reactor scale. The extent of synergy between phototrophy and heterotrophy is quantified using the carbon assimilated from both carbon uptake routes. The Pearson co-relation coefficient (Preuer et al., 2018) between two series of data is used here to quantify the synergy between the autotrophic and heterotrophic uptake of carbon. The carbon assimilation from individual trophic levels are obtained from eqns.(4)–(6), and the inter-trophic synergy is quantified as

$$\text{Pearson correlation coefficient}(r) = \frac{n(\Sigma xy) - (\Sigma x)(\Sigma y)}{\sqrt{[n\Sigma x^2 - (\Sigma x)^2][n\Sigma y^2 - (\Sigma y)^2]}} \quad (7)$$

where x = autotrophic C-assimilated calculated using eqn. (6), y = heterotrophic C-assimilated calculated using eqn. (5), and n = number of

data-sets (=6 for 10,500 lx, and 5 each for 12,000 and 14,000 lx).

****Fig. 6.j-l** compares the inter-trophic synergy (r) between the radially-mixed reactor and the control configuration under nitrogen-sufficient (NA) and limited (NL) conditions at optimal (10,500 lx) and super-optimal (12,000–14,000 lx) illumination. Radial mixing is observed to amplify inter-trophic in most cases (Fig. 6.j-l), and this increase in inter-trophic synergy is further amplified if 2 days of NL follows 4 days of NA. By imparting the algal cells with transverse momentum that fuels their frequent roundtrips between the well-lit peripheral region and inner darker regions and helps them recuperate from photo-damage from super-optimal light, radial mixing promotes the rapid conversion of solar energy to chemical energy (ATP and NADPH) that drive the light-independent reactions in both the phototrophic and the heterotrophic pathways. On the flip side, when rapid mixing reduces the extracellular pH below 7.5, both CO₂ sequestration and acetate assimilation are negatively impacted, thus affecting both the pathways concurrently (Cheng et al., 2022; Moheimani, 2013). Post the fourth day, when nitrogen limitation channels the growth towards lipid synthesis, the inter-trophic synergy increases since both the autotrophic and heterotrophic pathways focus on one target molecule, i.e., lipid. In the absence of nitrogen sources, the new influx of C-atoms do not have to compete with N-atoms for carbonaceous skeleton based substrates. Radial mixing further amplifies this synergy (Fig. 6.k-l) by homogenizing the inorganic (dissolved CO₂) and the organic (acetate) carbon across the reactor-section and ensuring that the carbon sources are available for the light-independent reactions in phototrophy and heterotrophy, regardless of the radial coordinates of the algal cells in the PBR.

4. Conclusions

This experimental and CFD study demonstrates how radial mixing simulates flashlight effect through transverse shuttling of microalgal cells in bubble-column PBRs, which promotes mixotrophic growth by enhancing synergy between phototrophy and heterotrophy, and reducing photo-limitation and photo-inhibition in the dark and the light zones, respectively. The inter-trophic synergy – which is quantified at varying illumination – increases phototrophic CO₂ sequestration (by 23–72 %) and acetate uptake from wastewater. A strategic combination of reactor-scale parameters – super-optimal CO₂ (6 %) and light (14,000 lx) with radial mixing and nitrogen limitation – in mass-scale outdoor algal PBRs would facilitate simultaneous bioremediation of CO₂-rich flue-gases and acetate-rich industrial wastewater.

CRediT authorship contribution statement

Gourab Ghosh: Writing – original draft, Visualization, Software, Methodology, Investigation, Formal analysis, Data curation. **Arnab Atta:** Writing – review & editing, Validation, Supervision, Resources. **Saikat Chakraborty:** Writing – original draft, Validation, Supervision, Resources, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.biortech.2025.132751>.

Data availability

Data will be made available on request.

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